

Year 10

Investigation 1



Baldivis
Secondary College

Surface Area and Volume

SOLUTIONS

Total Mark: _____

Name: _____

Class: _____

Total Time: **50** minutes

Equipment: *The Take-Home section of this investigation, INB Book, text book, Scientific Calculator*

Allocated marks

IN-CLASS VALIDATION: _____/37 Marks

*** START OF IN CLASS OPEN BOOK SECTION***

Introduction

In the in-class validation section of this investigation you are going to work through several calculations and designs to prove or disprove the following statement:

STATEMENT #2:

"The total surface area of a rectangular prism is always greater than its volume".

The calculations you completed during the take-home component of this investigation will help you to collect evidence to make an informed decision on whether the statement above is correct or not.

Take note that the method for finding the total surface area of a rectangular prism is:

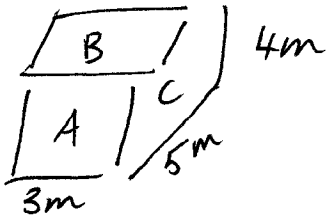
1. Label the length, width, and height of your rectangular prism.
2. Look at the six faces of the prism.
3. Find the area of the bottom face.
4. Find the area of the top face.
5. Find the area of the front and back faces.
6. Find the area of the left and right faces.
7. Add the six areas together.

To find the volume of a rectangular prism, we multiply the length of the prism by the width of the prism by the height of the prism.

Shape 1 calculations and evidence

Sketch and label a rectangular prism with the dimensions of a 3m x 5m x 4m.

[1 mark]



$\frac{1}{2}$ mark for sketch

$\frac{1}{2}$ mark for dimensions with units.

Calculate the surface area for this rectangular prism?

[4 marks]

$$\begin{aligned} \text{S.A of A} &= L \times W \\ &= 3 \times 4 \\ &= 12 \text{ m}^2 \end{aligned}$$

①

$$\begin{aligned} \text{S.A of B} &= L \times W \\ &= 3 \times 5 \\ &= 15 \text{ m}^2 \end{aligned}$$

①

$$\begin{aligned} \text{S.A of C} &= L \times W \\ &= 5 \times 4 \\ &= 20 \text{ m}^2 \end{aligned}$$

①

$$\begin{aligned} \text{T.S.A} &= (12 + 15 + 20) \times 2 \\ &= 94 \text{ m}^2 \end{aligned}$$

①

Calculate the volume for this rectangular prism?

[2 marks]

$$\begin{aligned} V &= L \times W \times h \\ &= 3 \times 5 \times 4 \\ &= 60 \text{ m}^3 \end{aligned}$$

Looking back at your surface area and volume calculations for this 3m x 5m x 4m rectangular prism, write 1-2 sentences proving or disproving the statement.

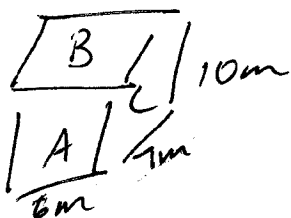
[1mark]

A rectangular prism with the dimensions of 3m x 5m x 4m has a surface area that is larger than its volume. So these calculations prove statement #2.

Shape 2 calculations and evidence

Sketch and label a rectangular prism with the dimensions of a 6m x 7m x 10m.

[1 mark]



see mark breakdown
from shape 1

Calculate the surface area for this rectangular prism?

[4 marks]

$$\begin{aligned} \text{S.A. of A} &= L \times W \\ &= 6 \times 10 \\ &= 60 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{S.A. of B} &= L \times W \\ &= 6 \times 7 \\ &= 42 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{S.A. of C} &= L \times W \\ &= 7 \times 10 \\ &= 70 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{T.S.A} &= (60 + 70 + 42) \times 2 \\ &= 344 \text{ m}^2 \end{aligned}$$

see mark breakdown
from shape 1

Calculate the volume for this rectangular prism?

[2 marks]

$$\begin{aligned} V &= L \times W \times h \\ &= 6 \times 7 \times 10 \\ &= 420 \text{ m}^3 \end{aligned}$$

see mark breakdown
from shape 1

Looking back at your surface area and volume calculations for this 6m x 7m x 10m rectangular prism, write 1-2 sentences proving or disproving the statement.

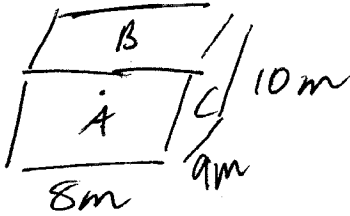
[1 mark]

①
Volume is larger than S.A for
this sized rectangular prism.

Shape 3 calculations and evidence

Sketch and label a rectangular prism with the dimensions of a 8m x 9m x 10m.

[1 mark]



See mark break
down from shape 1

Calculate the surface area for this rectangular prism?

[4 marks]

$$\begin{aligned} \text{S.A of A} &= L \times W \\ &= 8 \times 10 \\ &= 80 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{S.A of B} &= L \times W \\ &= 8 \times 9 \\ &= 72 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{S.A of C} &= L \times W \\ &= 9 \times 10 \\ &= 90 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{T.S.A} &= (80 + 72 + 90) \times 2 \\ &= 484 \text{ m}^2 \end{aligned}$$

See mark down
from shape 1

Calculate the volume for this rectangular prism?

[2 marks]

$$\begin{aligned} V &= L \times W \times h \\ &= 8 \times 9 \times 10 \\ &= 720 \text{ m}^3 \end{aligned}$$

See mark break down
from shape 1

Looking back at your surface area and volume calculations for this 8m x 9m x 10m rectangular prism, write 1-2 sentences proving or disproving the statement.

[1mark]

①
Volume is larger than
Surface area

Summary: Write a paragraph that summarizes the findings of all 3 different sized rectangular prisms.

[4 marks]

Your summary should include:

- What you are answering
- Re-stating your calculations
- What did you find out?
- What is your opinion of the statement

See example from the take-home
Section.

Additional calculations and evidence

STATEMENT #3:

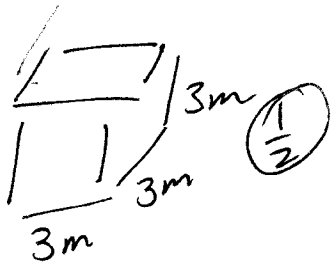
"If you combined 2 cubes to form a rectangular prism, does the volume and surface area double?"

[9 marks]

After reading statement #3, use examples to calculate and explain your position towards this statement.

You will need to use a similar method as you did for statement 1 and 2 (i.e. sketching and labelling, finding the surface areas, finding the volumes and summarising your findings)

The example I will use is a $3 \times 3 \times 3$ cube.



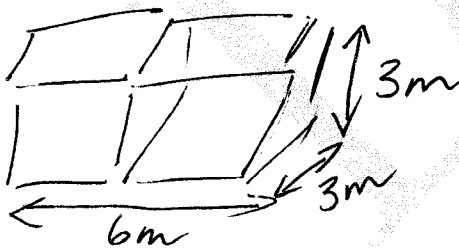
$$\begin{aligned} S.A. &= (L \times L) \times 6 \\ &= 3 \times 3 \times 6 \\ &= 54m^2 \end{aligned}$$

$$\begin{aligned} V &= L \times L \times h \\ &= 3 \times 3 \times 3 \\ &= 27m^3 \end{aligned}$$

If the volume + surface area is doubled to form a rectangular prism then the

$$\begin{aligned} S.A. &= 54 \times 2 \\ &= 108m^2 \\ V &= 27 \times 2 \\ &= 54m^3 \end{aligned}$$

Sketch of 2 cubes joined together



$$\begin{aligned} S.A. &= 2 \times (6 \times 3) + 2 \times (3 \times 3) + 2 \times (6 \times 3) \\ &= 2 \times 18 + 2 \times 9 + 2 \times 18 \\ &= 36 + 18 + 36 \\ &= 90m^2 \end{aligned}$$

$$\begin{aligned} Vol &= L \times W \times h \\ &= 6 \times 3 \times 3 \\ &= 54m^3 \end{aligned}$$

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These calculations show that whilst the volume doubles the surface area does not. That is because you lose some of the faces (surfaces) when the cubes join.

DRAFT

*** END OF IN-CLASS SECTION***